

assignment energy_work

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:03pm

Quiz Instructions

No video. But please, use a post-it to hide the hints. Try without looking the hints.



Question 1 6 pts

A traveler pulls on a suitcase strap at an angle 36° above the horizontal. If 555J of work are done by the strap while moving the suitcase a horizontal distance of 15m what is the tension in the strap?

hint: work = (Fx) (displacement x) Fx is the x-component along x-axis



56 N



37 N



46 N



52 N



Question 2 6 pts

What is the minimum energy needed to change the speed of a 1600-kg sport utility vehicle from 15.0 m/s to 40.0 m/s?

hint: find the change in KE. M means Mega (1 million), K means Kilo (1000)



40.0 kJ



10.0 kJ



0.960 MJ



20.0 kJ



1.10 MJ



Question 3 6 pts

Find the net work done by friction on a box that moves in a complete circle of radius 1.82m on a uniform horizontal floor. The coefficient of kinetic friction between the floor and the box is 0.25, and the box weighs 65N

hint: friction is a nonconservative force. distance matters. work = distance x frictional force. Here the distance is the circumference. frictional force = μ Normal. And remember Normal = weight (up=down)

☐

0 J

☐

190 J

☐

1800 J

☐

370 J



Question 4 6 pts

A 10-kg dog is running with a speed of 5.0 m/s. What is the minimum work required to stop the dog in 2.40 s?

hint: its a tricky question because work does not depend on the time. Work is the change in KE. So find the change in KE. The final velocity is 0 because it stops.

☐

100 J

☐

75 J

☐

125 J

☐

50 J



Question 5 6 pts

The work you do when pushing a shopping cart twice as far while applying the same force is

☐

twice as much.

☐

four times as much.

☐

half as much.



the same amount.



Question 6 6 pts

How large a net force is required to accelerate a 1600-kg SUV from rest to a speed of 25 m/s in a distance of 200 m?

hint: work is the change in KE. and initial velocity is 0.

Find the work then work is force x distance.



0 N



400 N



2500N



1600 N



200 N



Question 7 6 pts

An object has gravitational potential energy due to its



location.



momentum.



none of the above



acceleration.



speed.



Question 8 6 pts

A 1.0-kg object moving in a certain direction has a kinetic energy of 2.0 J. It hits a wall and comes back with half its original speed. What is the kinetic energy of this object at this point?

Hint: first use the $KE=2J$ to find the initial speed. Then divide that speed by 2 and compute the new KE.



0.50 J



1.0 J



2.0 J



0.25 J



4.0 J

**Question 9 6 pts**

A 2-kg ball is held 4 m above the ground. Relative to the ground its potential energy is



more than 80 J.



6 J.



80 J.



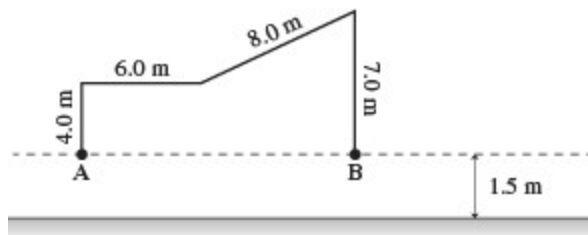
8 J.



32 J.

**Question 10 6 pts**

A person carries a 25.0-N rock through the path shown in the figure, starting at point A and ending at point B. The total time from A to B is 1.50 min. How much work did gravity do on the rock between A and B?



hint: weight is a conservative force. Work depends only on $(h_B - h_A)$. difference i height.



75 J



20.0 J



0 J



275 J



625 J

**Question 11 6 pts**

The work done in pushing a TV set a distance of 2 m with an average force of 20 N is

☐

800 J.

☐

40 J.

☐

10 J.

☐

20 J.

☐

2 J.

**Question 12 6 pts**

When a car of mass 1167kg accelerates from 10.0 m/s to some final speed, 4.00×10^5 J of work are done. Find this final speed.

hint: Work (J) is the change in KE. $W = 0.5 \text{ mass } (V_f^2 - V_i^2)$ Solve for V_f .

☐

28.0 m/s

☐

25.2 m/s

☐

22.4 m/s

☐

30.8 m/s

**Question 13 6 pts**

A stone with a mass of 1.0 kg is tied to the end of a light string which keeps it moving in a circle with a constant speed of 4.0 m/s on a perfectly smooth horizontal tabletop. The radius of the path is 0.60 m. How much work does the tension in the string do on the stone as it makes one-half of a complete circle?

hint: remember. A force is doing work if it has a component along the displacement (trajectory). If a force or component is perpendicular to the displacement, there is no work done (does not help increasing the KE).

☐

0 J

☐

3.8 J

☐

100 J

☐

80 J

☐

40 J



Question 14 6 pts

How much kinetic energy does a 0.30-kg stone have if it is thrown at 44m/s

☐

510 J

☐

290 J

☐

440 J

☐

580 J



Question 15 6 pts

The kinetic friction force that a horizontal surface exerts on a 60.0-kg object is 50.0 N. If the initial speed of the object is 25.0 m/s, what distance will it slide before coming to a stop?

hint: force find the work. Work is the change in KE. Final speed is 0. Then Work is force x distance and frictional force is given.

☐

15.0 m

☐

30.0 m

☐

750 m

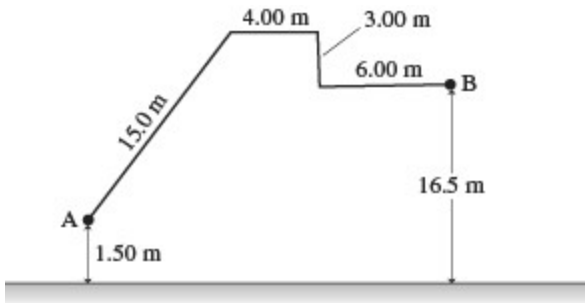
☐

375 m



Question 16 6 pts

A person carries a 2.00-N pebble through the path shown in the figure, starting at point A and ending at point B. The total time from A to B is 6.75 min. How much work did gravity do on the rock between A and B?



hint: weight is a conservative force. The work depends only on the difference in height. If you bring the stone from A to B. straight up. Is gravity doing positive or negative work? Its positive if it opposes the displacement (trying to slow down the stone).

☐

30.0 J

☐

-36.0 J

☐

-56.0 J

☐

-30.0 J

☐

56.0 J

☐

Question 17 6 pts

If a sprinter running at a speed of 10 meters per second could convert his/her kinetic energy into upward motion, how high could he/she jump? ($KE = PE$ so $0.5mV^2 = mgh$. cross m)

☐

10m

☐

1m

☐

100m

☐

5m

Not saved

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